

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			transfer of one electron from calcium to each chlorine atom (1) can be implied by electrons being in correct place in chloride ions octet in each chloride ion (1) ignore octet included in calcium ion correct charge on all ions – Ca ²⁺ and Cl ⁻ (1)		3		3		
	(b)			 four shared pairs (1) octet around Cl atom (1)		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Indicative content calcium has metallic bonding / is a metal delocalised/free electrons between positive ions electrons move towards positive terminal electric current is the movement of electrons / electrons carry electricity calcium chloride has ionic bonding / is an ionic substance solid cannot conduct electricity because ions cannot move ions become free when calcium chloride is molten or in solution charged ions travel towards oppositely charged electrodes 5-6 marks Good understanding of the different structures and that electrons move in metals whilst ions move when ionic compounds are melted/dissolved <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Basic understanding of one of the structures and idea that electrons/ions move when a substance conducts electricity <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Some knowledge of metallic or ionic bonding and structure <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	6			6		
				Question 4 total	6	5	0	11	0	0